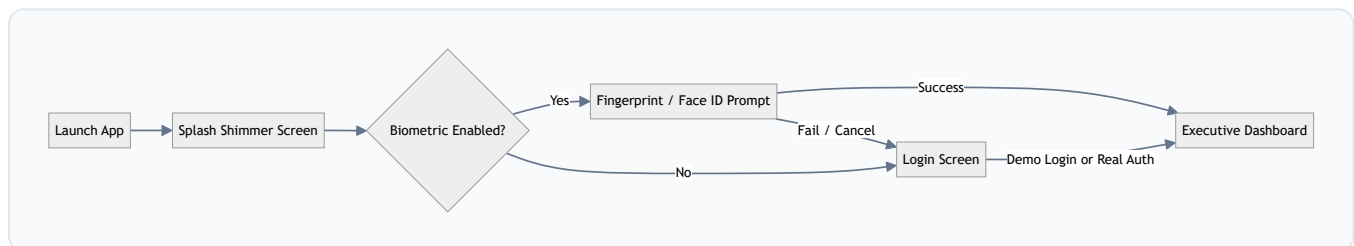


SecurePulse Mobile — End-to-End Walkthrough & Operations Guide

This guide provides a comprehensive walkthrough of the **SecurePulse Mobile** application for security analysts, incident responders, administrators, and developers.

1. Application Launch & Authentication



Steps:

- Splash Screen:** On cold boot, the app displays the SecurePulse holographic cyber shield with smooth entrance animations and initializes the encrypted Hive storage, Push notification handlers, and cached user preferences.
- Biometric Challenge:** If enabled in settings, the OS biometric gate prompts for instant authentication.
- Login Options:**
 - **Demo Login:** Tap "Enter as Security Analyst (Demo Mode)" for instant access without network setup.
 - **Production Login:** Enter enterprise credentials (`email` and `password`) to obtain an authenticated JWT from the FastAPI backend.

2. Executive Security Dashboard (Tab 1)

The home dashboard is an executive-level overview of real-time organizational security posture.

Key Components:

- **Security Posture Score:** Circular animated gauge displaying the composite security score (e.g., **94/100 • Grade A+**).
- **Live System Status Badges:**
 - `Cloud API` : Connection status to Render FastAPI server.
 - `WebSocket` : Real-time incident stream status.
 - `GitHub SAST` : Codebase sync status.

- **AI Engine** : LLM security assistant readiness.
- **Vulnerability Breakdown Chart**: Interactive `fl_chart` donut graph categorizing Critical, High, Medium, and Low vulnerabilities.
- **Quick Action Dock**:
 -  *Trigger SAST Scan*
 -  *SOC Alert Triage*
 -  *Ask AI Security Copilot*
 -  *Export Compliance PDF*

3. Repositories & SAST Security Audits (Tab 2)

Monitors enterprise source code repositories for security vulnerabilities, secrets leakage, and misconfigurations.

Workflow:

1. **Repository List**: Displays repositories (e.g., `secureguard-mobile`, `secureguard-backend`, `payment-gateway`).
2. **Repository Detail**:
 - Primary language, branch, commit SHA, and overall security rating.
 - Scan history with timestamps and finding tallies.
 - **"Trigger Immediate SAST Scan"** button: Initiates Semgrep static analysis across the codebase.
3. **Findings Drilldown**:
 - Filter by severity: **Critical, High, Medium, Low**.
 - Tap any finding to view the exact code snippet, line number, CVE reference, CWE classification, and remediation guidance.

4. AI Security Copilot (`SecurePulse AI`) (Tab 3)

An interactive conversational assistant trained on cybersecurity frameworks (SOC 2 Type II, ISO/IEC 27001, PCI-DSS, NIST CSF) and incident response playbooks.

Sample Prompts & Capabilities:

- **CVE Explanation**: *"Explain CVE-2024-3094 and provide the patch instructions for Debian/Ubuntu."*
 - **Remediation Code**: *"Generate a secure FastAPI CORS policy with strict origin validation."*
 - **Firewall Rules**: *"Write iptables and ufw rules to mitigate DDoS SYN flood on port 443."*
 - **Compliance Queries**: *"What are the required controls for SOC 2 Type II CC6.1 logical access controls?"*
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⚡ 5. SOC Threat Stream & Incident Alerts (Tab 4)

Real-time SIEM incident management powered by WebSockets.

Alert Investigation Workflow:

1. **Live Incident Feed:** Chronological list of real-time security events (e.g., *Brute Force Detection on SSH, SQL Injection Attempt, Unauthorized IAM Privilege Escalation*).
 2. **Filtering:** Instant filtering by severity level and status (Active, In Progress, Resolved).
 3. **Incident Detail View:**
 - Source: Wazuh SIEM, CloudTrail, or Snort IDS.
 - Attacker IP, Destination Port, and Geographic Origin.
 - Full JSON payload and raw syslog stream.
 - Action buttons: **"Acknowledge Alert"**, **"Quarantine IP"**, and **"Generate Incident Report"**.
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📄 6. Executive Compliance Reports & PDF Export

Generate cryptographically signed compliance reports directly to the mobile file system or share sheet.

Supported Frameworks:

- **SOC 2 Type II** (Trust Services Criteria CC6.1 - CC8.1)
- **ISO/IEC 27001:2022** (Information Security Management)
- **PCI-DSS v4.0** (Payment Card Industry Security Standards)
- **HIPAA Security Rule** (Healthcare Protected Health Information)

Export Features:

- Cryptographic SHA256 audit fingerprint included on every page.
 - One-tap export via Android/iOS Share Sheet (Email, Slack, WhatsApp, Google Drive, AirDrop).
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⚙️ 7. Settings & Environment Diagnostics (Tab 5)

Complete diagnostic and configuration control center.

Configuration Controls:

- **Environment Switcher:**
 - 🌐 *Render Cloud* (<https://secureguard-backend-7eqm.onrender.com>)
 - 📱 *Android Emulator Loopback* (<http://10.0.2.2:8000>)
 - 💻 *Localhost* (<http://127.0.0.1:8000>)
 - 🔧 *Custom URL Input*
- **Demo Mode Toggle:** One-tap switch between offline simulation and live backend.
- **Biometric Auth Toggle:** Turn Face ID / Fingerprint lock on or off.

- **Theme Selector:** Toggle between Cyber Dark and Clean Light modes.
- **Network Diagnostics:** Real-time HTTP ping test and WebSocket handshake validation.